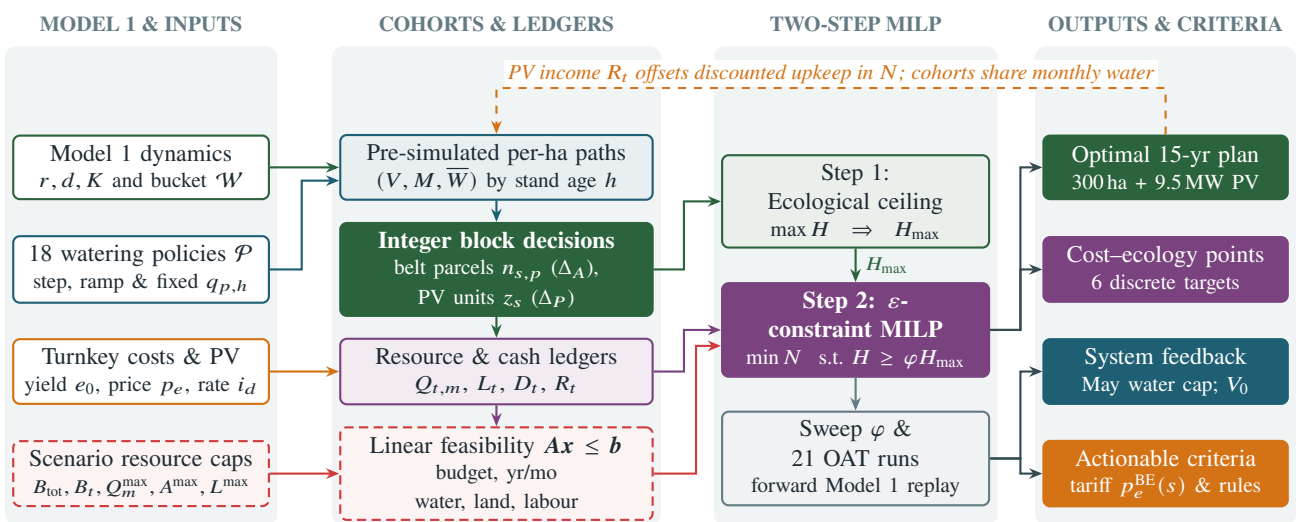

MODEL 1 & INPUTS
COHORTS & LEDGERS
TWO-STEP MILP
OUTPUTS & CRITERIA
PV income R_t offsets discounted upkeep in N; cohorts share monthly water
Model 1 dynamics
r, d, K and bucket $\mathcal{W}$
18 watering policies $\mathcal{P}$
step, ramp & fixed $q_{p,h}$
Turnkey costs & PV
yield e_0, price p_e, rate i_d
Scenario resource caps
$B_{\text{tot}}, B_t, Q_m^{\max}, A^{\max}, L^{\max}$
Pre-simulated per-ha paths
$(V, M, \overline{W})$ by stand age h
Integer block decisions
belt parcels $n_{s,p}$ (Δ_A),
PV units z_s (Δ_P)
Resource & cash ledgers
$Q_{t,m}$, L_t, D_t, R_t
Linear feasibility $\boldsymbol{Ax} \le \boldsymbol{b}$
budget, yr/mo
water, land, labour
Step 1:
Ecological ceiling
max H ⇒ $H_{\max}$
$H_{\max}$
Step 2: ε-constraint MILP
min N s.t. $H \ge \varphi H_{\max}$
Sweep φ &
21 OAT runs
forward Model 1 replay
Optimal 15-yr plan
300 ha + 9.5 MW PV
Cost–ecology points
6 discrete targets
System feedback
May water cap; V_0
Actionable criteria
tariff $p_e^{\text{BE}}(s)$ & rules